

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Deep Learning

COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 5

Duration: 1 Hour

Total Marks:50

Annexure A

Analytical Problem Solving

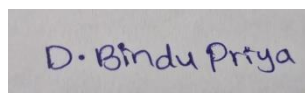
1. A coin is tossed 10 times and the outcomes are:
H H T H H H T H H T
Using Maximum Likelihood Estimation, estimate the probability of getting Heads (p).
2. Out of 50 students, 42 passed an exam.
Assuming a Bernoulli distribution, find the MLE of the probability that a student passes.
3. A factory produces light bulbs. In a sample of 100 bulbs, 8 are defective.
Find the MLE for the probability that a bulb is defective.
4. A biased die is rolled 60 times. The number 6 appears 18 times.
Estimate the probability of rolling a 6 using MLE.
5. Given:
Input (x) = 4
Weight (w) = 2
Actual Output = 10
Prediction:
 $y = w \times x$
 1. Find the predicted output.
 2. Calculate the error.
 3. If the gradient is 4 and learning rate is 0.01, find the updated weight.

Code of Conduct:

I D.BINDU PRIYA(192425162) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

1. A Coin is Tossed 10 Times

Problem Understanding

A coin is tossed 10 times, producing the following sequence:

H H T H H H T H H T

The objective is to estimate the probability of obtaining a Head using the **Maximum Likelihood Estimation (MLE)** method. MLE estimates the parameter that makes the observed data most likely to occur. Since each toss has only two outcomes (Head or Tail), this follows a Bernoulli/Binomial model.

Method Selection

For Bernoulli or Binomial data, the Maximum Likelihood Estimator of the probability of success is

$$\hat{p} = \frac{x}{n}$$

where

- x = Number of successful outcomes (Heads)
- n = Total number of trials

This method is selected because it provides the most reliable estimate of the unknown probability based on the observed sample.

Solution Process

Step 1: Count the number of Heads.

$$\text{Heads} = 7$$

Step 2: Count the total number of tosses.

$$\text{Total tosses} = 10$$

Step 3: Apply the MLE formula.

$$\hat{p} = \frac{7}{10} = 0.7$$

Accuracy of Calculation

Number of Heads = 7

Total Tosses = 10

Estimated Probability

$$\hat{p}=0.7$$

Interpretation of Results

The Maximum Likelihood Estimate of the probability of obtaining a Head is **0.7 (70%)**. This means that based on the observed sample, the coin produced Heads approximately 70% of the time. Therefore, the coin appears to be biased toward Heads in this experiment.

2. Students Passing an Examination

Problem Understanding

A class contains **50 students**, and **42 students passed** the examination. The task is to estimate the probability that a randomly selected student passes the examination using Maximum Likelihood Estimation.

Since each student either **passes or fails**, the data follow a Bernoulli distribution.

Method Selection

The Maximum Likelihood Estimator for the probability of success is

$$\hat{p}=\frac{x}{n}$$

where

- x = Number of students who passed
- n = Total number of students

Solution Process

Step 1: Number of students who passed

$$x=42$$

Step 2: Total students

$$n=50$$

Step 3: Calculate the estimate

$$\hat{p} = \frac{42}{50} = 0.84$$

Accuracy of Calculation

Estimated probability

$$\hat{p} = 0.84$$

Interpretation of Results

The estimated probability of passing is **0.84 (84%)**. This indicates that approximately 84 out of every 100 students are expected to pass under similar conditions.

3. Defective Light Bulbs

Problem Understanding

A factory inspects **100 light bulbs**, and **8 bulbs are found to be defective**. The aim is to estimate the probability that a manufactured bulb is defective using Maximum Likelihood Estimation.

Each bulb has two possible outcomes:

- Defective
- Non-defective

Hence, the Bernoulli distribution is applicable.

Method Selection

The MLE formula is

$$\hat{p} = \frac{x}{n}$$

where

- x = Number of defective bulbs
- n = Total bulbs inspected

Solution Process

Step 1: Defective bulbs

$$x=8$$

Step 2: Total bulbs

$$n=100$$

Step 3: Estimate probability

$$\hat{p} = \frac{x}{n} = \frac{8}{100} = 0.08$$

Accuracy of Calculation

Estimated probability

$$\hat{p} = 0.08$$

Interpretation of Results

The probability that a bulb is defective is **0.08 (8%)**. This indicates that the manufacturing process has a relatively low defect rate, with approximately 8 defective bulbs in every 100 produced.

4. Biased Die

Problem Understanding

A die is rolled **60 times**, and the number **6 appears 18 times**. The objective is to estimate the probability of obtaining a 6 using Maximum Likelihood Estimation.

Method Selection

MLE estimates the probability using

$$\hat{p} = \frac{x}{n}$$

where

- x = Number of times 6 occurred
- n = Total rolls

Solution Process

Step 1: Number of sixes

$$x=18$$

Step 2: Total rolls

$$n=60$$

Step 3: Apply the formula

$$\hat{p} = \frac{18}{60} = 0.30$$

Accuracy of Calculation

Estimated probability

$$\hat{p} = 0.30$$

Interpretation of Results

The estimated probability of rolling a **6** is **0.30 (30%)**. Since a fair die has a probability of approximately **0.167**, the estimated value suggests that the die is biased in favor of the number 6.

5. Prediction, Error Calculation and Weight Update

Problem Understanding

Given:

- Input $x=4$
- Weight $w=2$
- Actual Output = 10
- Prediction formula

$$y = w \times x$$

$$\text{Gradient} = 4$$

Learning Rate = 0.01

The task is to calculate the predicted output, determine the prediction error, and update the weight using the gradient descent algorithm.

Method Selection

The following equations are used:

Prediction

$$y = w \times x$$

Prediction Error

$$\text{Error} = \text{Actual Output} - \text{Predicted Output}$$

Weight Update Rule

$$w_{\text{new}} = w - \eta \times \text{Gradient}$$

where

$$\eta = 0.01$$

Solution Process

Step 1: Calculate Predicted Output

Given

$$w = 2, x = 4$$

Prediction

$$y = 2 \times 4 \quad y = 8$$

Step 2: Calculate Prediction Error

Actual Output

$$= 10$$

Predicted Output

=8

Therefore,

Error=10-8=2

Step 3: Update Weight

Using gradient descent,

$w_{\text{new}} = 2 - (0.01 \times 4)$ $w_{\text{new}} = 2 - 0.04$ $w_{\text{new}} = 1.96$

Accuracy of Calculation

Parameter	Value
Predicted Output	8
Error	2
Updated Weight	1.96

Interpretation of Results

The model predicts an output of **8**, while the actual output is **10**, resulting in an error of **2**. The gradient indicates the direction in which the weight should be adjusted to reduce the prediction error. After applying the gradient descent update rule with a learning rate of **0.01**, the weight changes from **2.00** to **1.96**. Repeating this process over multiple training iterations gradually minimizes the prediction error and improves the model's accuracy.